Slide 1: Title Slide

Title: Data Mining – Unlocking Value from Data
Subtitle: Techniques, Applications & Industry Insights

📌 Slide 2: What is Data Mining?

Definition:
Data Mining is the process of **extracting meaningful patterns**, **trends**, and **knowledge** from large datasets using statistical, machine learning, and database techniques.

Keywords: Pattern discovery, Prediction, Classification, Clustering

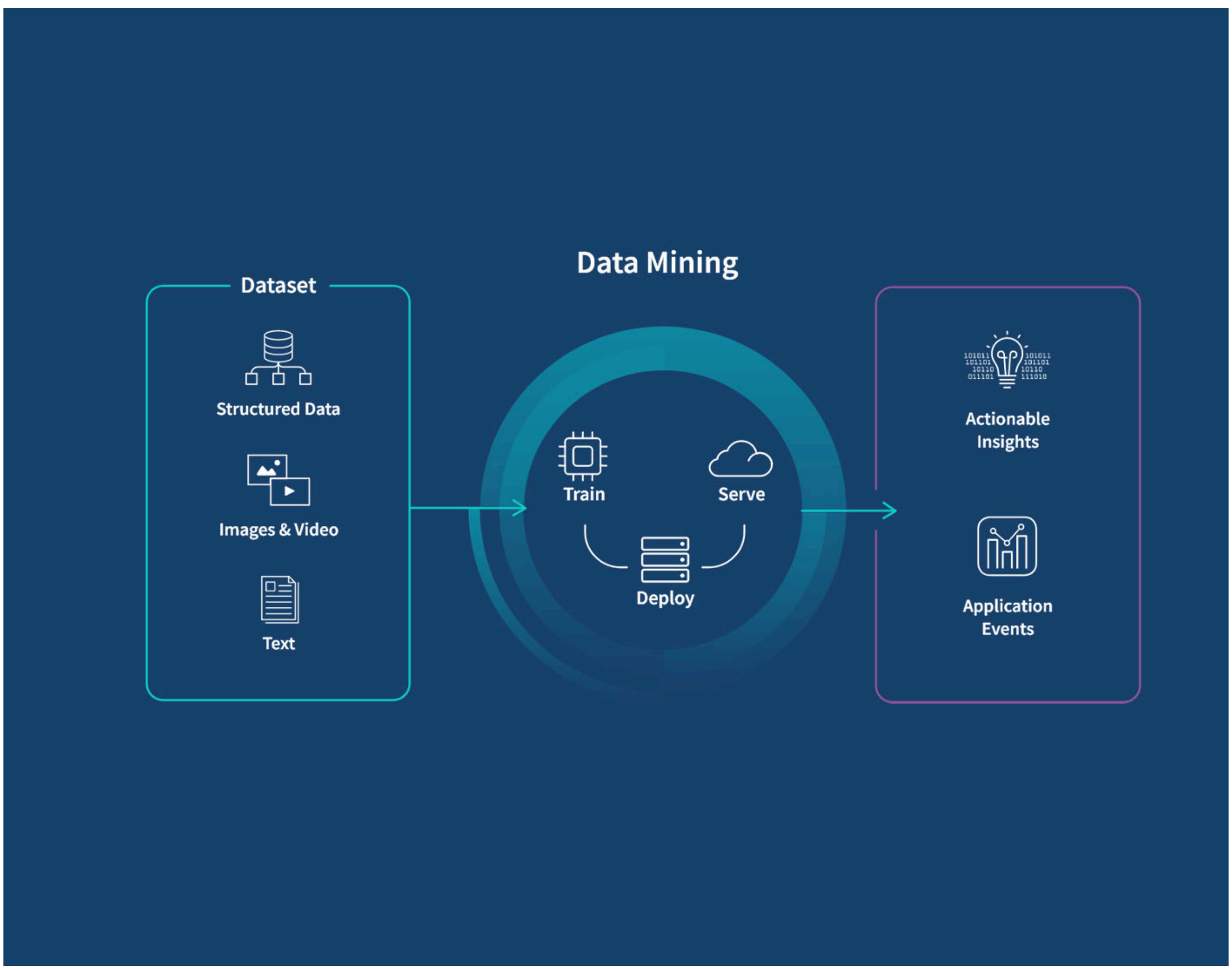


Illustration:

A funnel showing:
Raw Data → Processing → Patterns → Insight → Decision

🔄 Slide 3: The Knowledge Discovery Process (KDD)

Steps in KDD:

1. **Selection** – Choose relevant data
2. **Preprocessing** – Clean and integrate data
3. **Transformation** – Reduce and transform to suitable format
4. **Data Mining** – Apply algorithms to discover patterns
5. **Interpretation/Evaluation** – Present knowledge

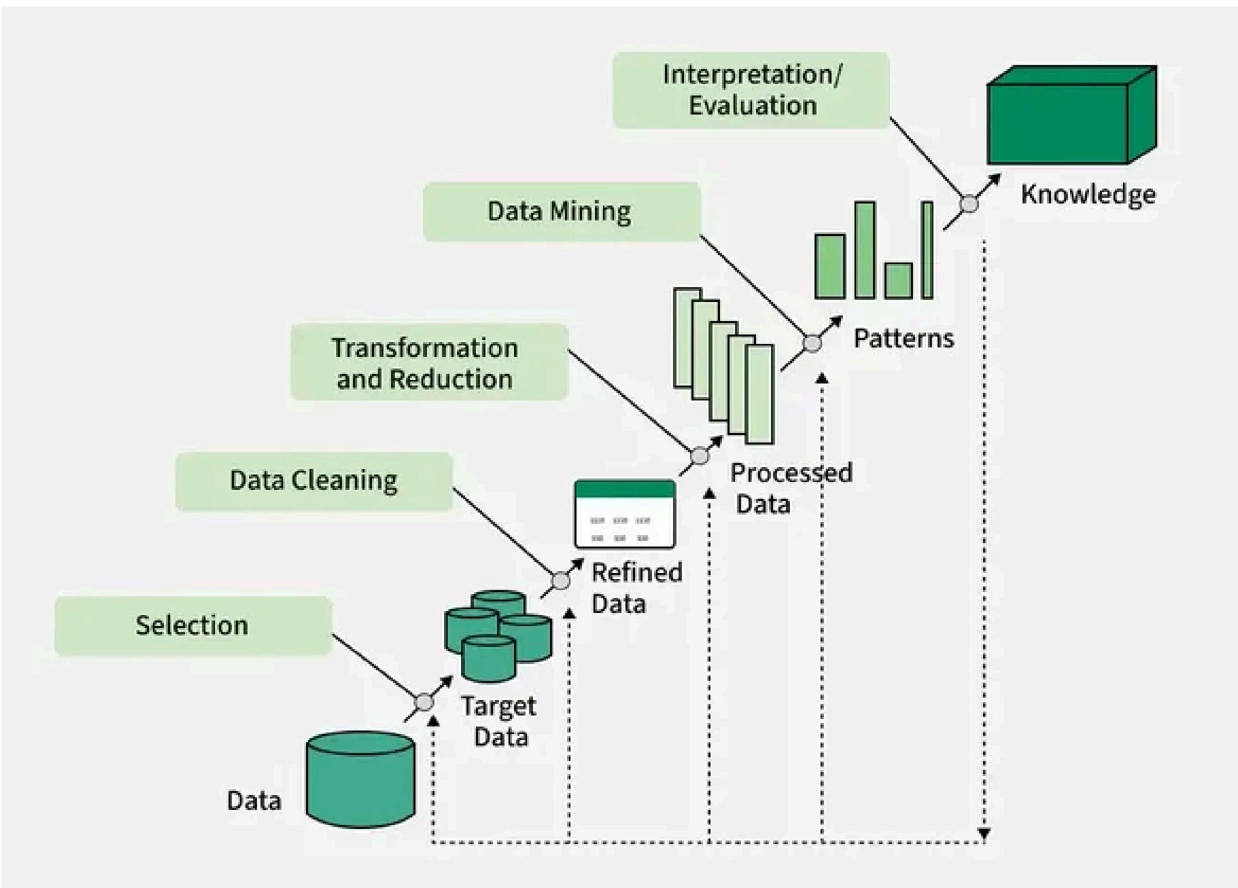


Illustration:
Flowchart showing KDD steps with arrows connecting boxes.

🧩 Slide 4: Need for Data Mining

Why Data Mining is Important:

- Growing data volumes (Big Data)
- Discovering **hidden patterns**
- Automating **decision-making**
- Supporting **business strategy**
- Enhanced **customer experience**

Example:
Retailers use data mining to find associations between purchases:

```sql
-- Market Basket Analysis Example
SELECT product_a, product_b, COUNT(*) AS frequency
FROM transactions
GROUP BY product_a, product_b
HAVING COUNT(*) > 100;
```

🎯 Slide 5: Uses of Data Mining

Domain	Purpose
Banking	Fraud detection, credit scoring
Retail	Market basket analysis, CRM
Healthcare	Disease prediction
Manufacturing	Quality control, forecasting
Education	Student performance prediction

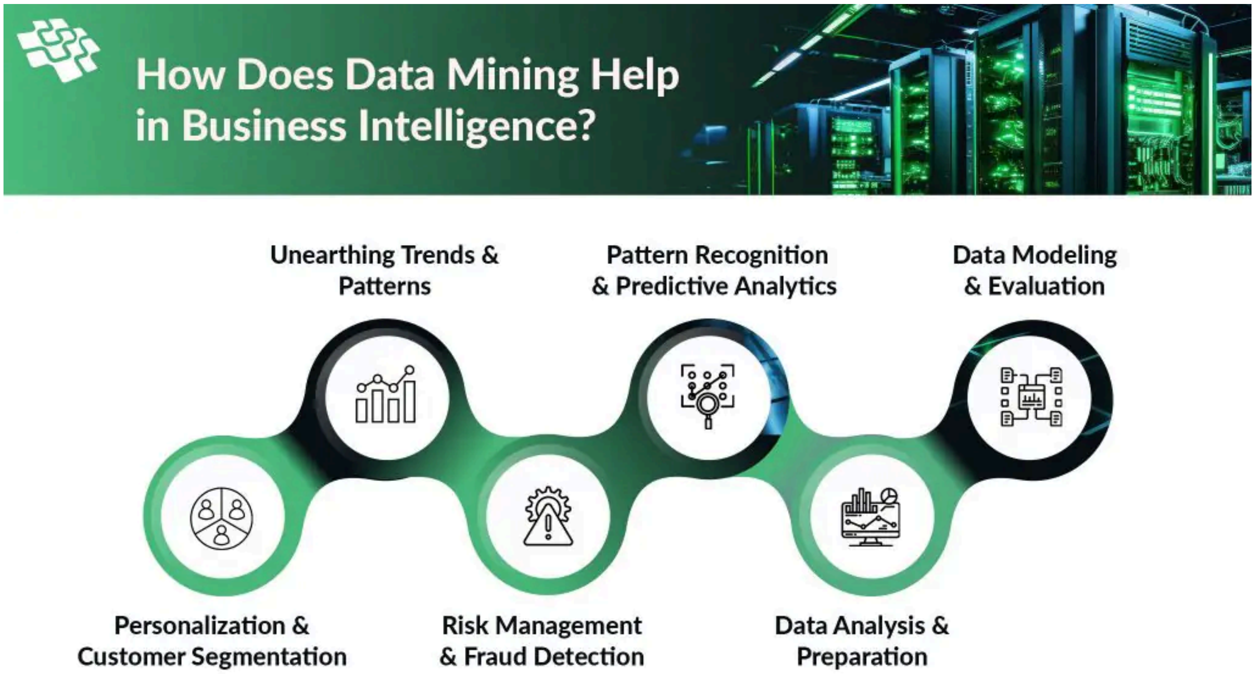
🤝 Slide 6: Data Mining + Business Intelligence

How They Work Together:

- **Data Mining** = Pattern Discovery
- **BI** = Reporting, Visualization, Decision Support

Combined Power:

- Better dashboards
- Real-time insights
- Predictive analysis


How Does Data Mining Help in Business Intelligence?
Unearthing Trends & Patterns
Pattern Recognition & Predictive Analytics
Data Modeling & Evaluation
Personalization & Customer Segmentation
Risk Management & Fraud Detection
Data Analysis & Preparation

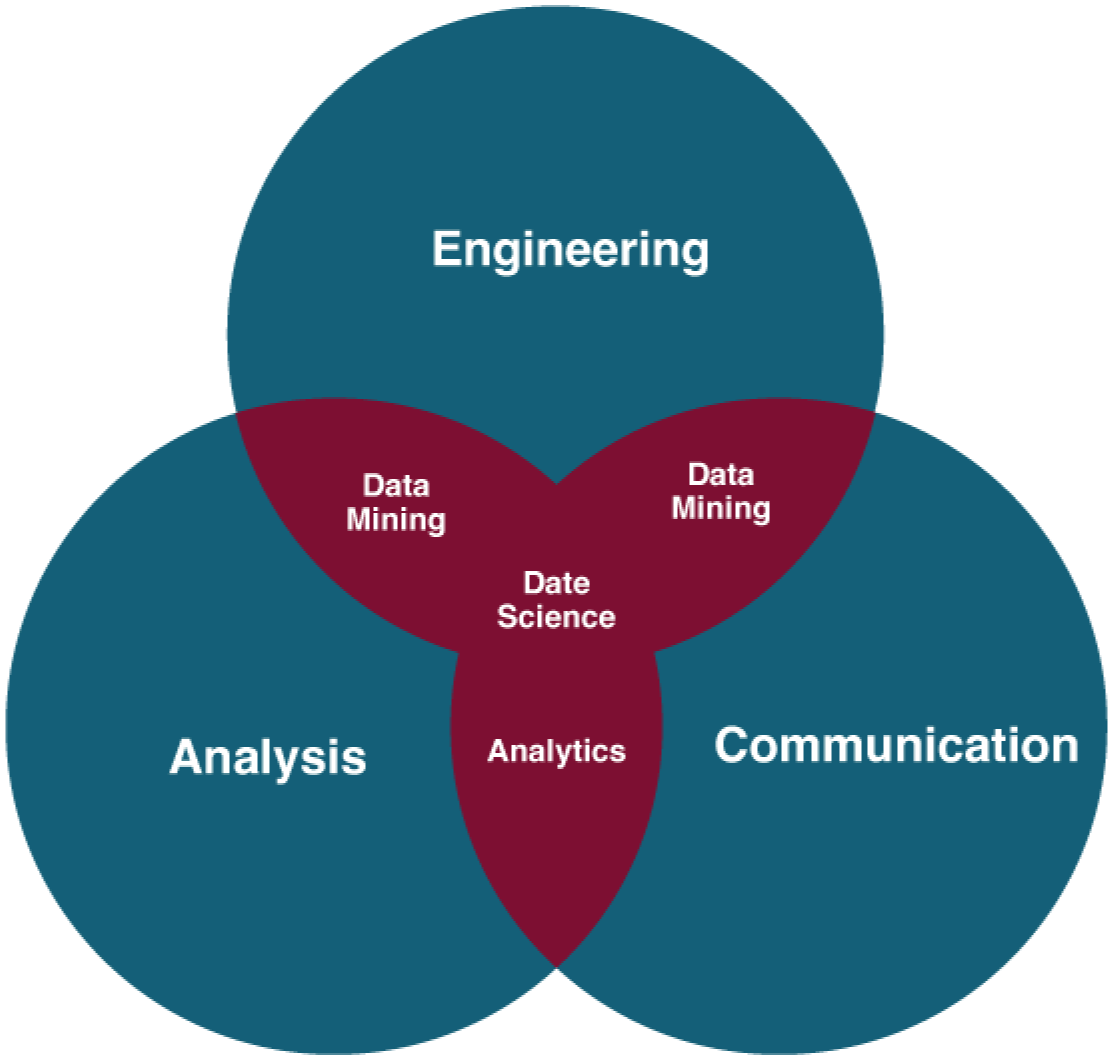


Illustration:
BI tools like Tableau/Power BI integrating with data mining results

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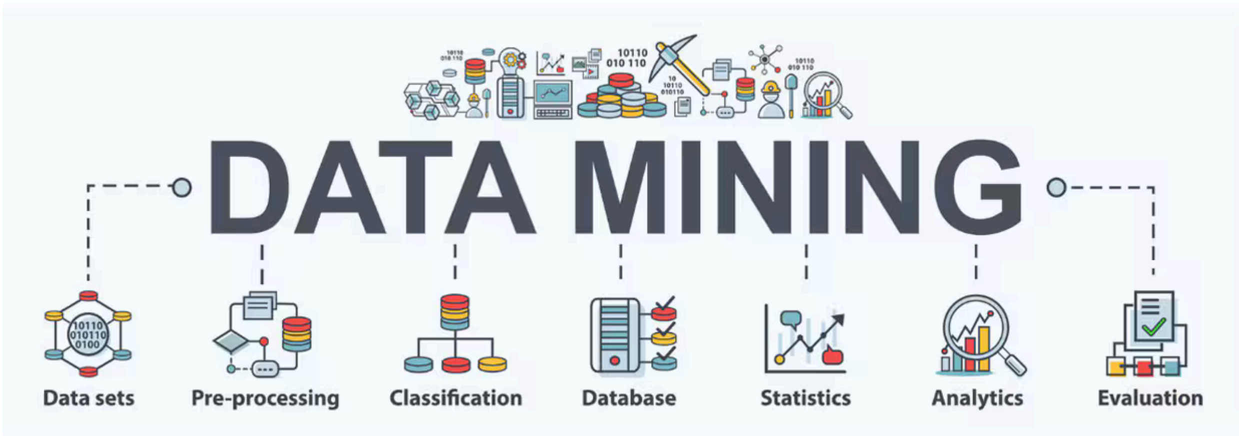
📂 Slide 7: Types of Data Used in Data Mining

1. **Relational Databases**
2. **Data Warehouses**
3. **Transactional Data**
4. **Text Data (NLP)**
5. **Spatial & Temporal Data**
6. **Multimedia Data (Images, Video)**
7. **Web Data (Logs, Clickstreams)**



Example (Text Mining):

```
from sklearn.feature_extraction.text import CountVectorizer
cv = CountVectorizer()
data = cv.fit_transform(["chatbot mining", "mining insights"])
```

🚀 Slide 8: Data Mining Applications

- **Customer Segmentation**
- **Market Basket Analysis**
- **Fraud Detection**
- **Churn Prediction**
- **Recommendation Systems**
- **Risk Management**
- **Web Mining**
- **Healthcare Diagnostics**

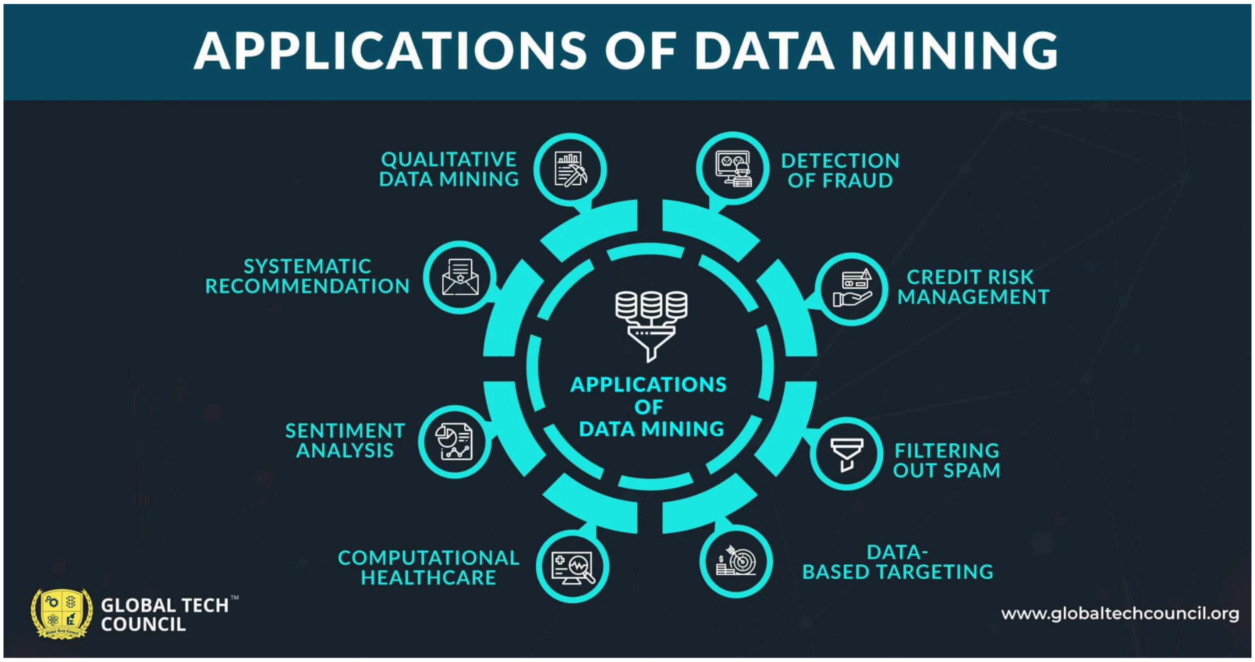


Illustration:
Icons representing each domain around a central data hub.

🧰 Slide 9: Popular Data Mining Products

Tool	Type	Vendor / Platform
RapidMiner	Open-source	RapidMiner Inc.
WEKA	Open-source	University of Waikato
SAS EM	Commercial	SAS
KNIME	Open-source	KNIME GmbH
IBM SPSS	Commercial	IBM
Oracle DM	Built-in	Oracle Database

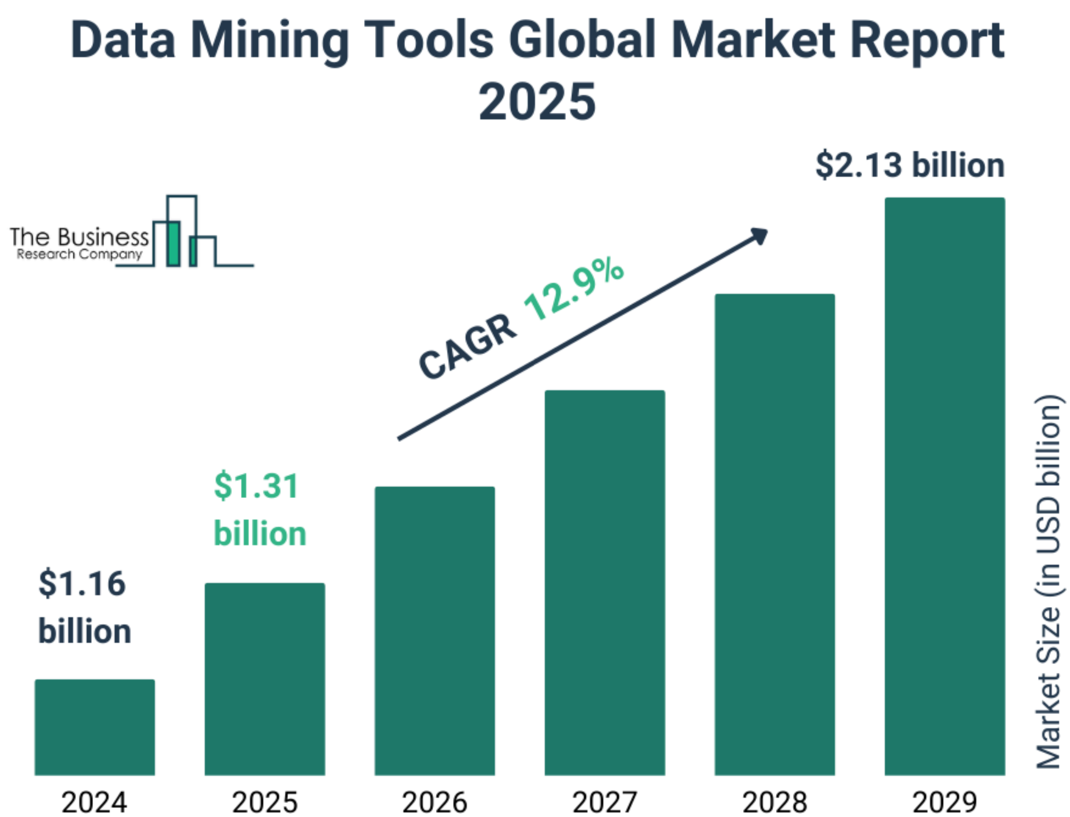
💹 Slide 10: Data Mining Market Trends

Market Insights:

- CAGR > 10% globally
- Rising adoption in **healthcare**, **retail**, and **finance**
- AI + Data Mining integration
- Cloud-based mining platforms

Forecast:

- Valued over $30B+ by 2030
- Growth driven by **Big Data**, **AI**, and **IoT**



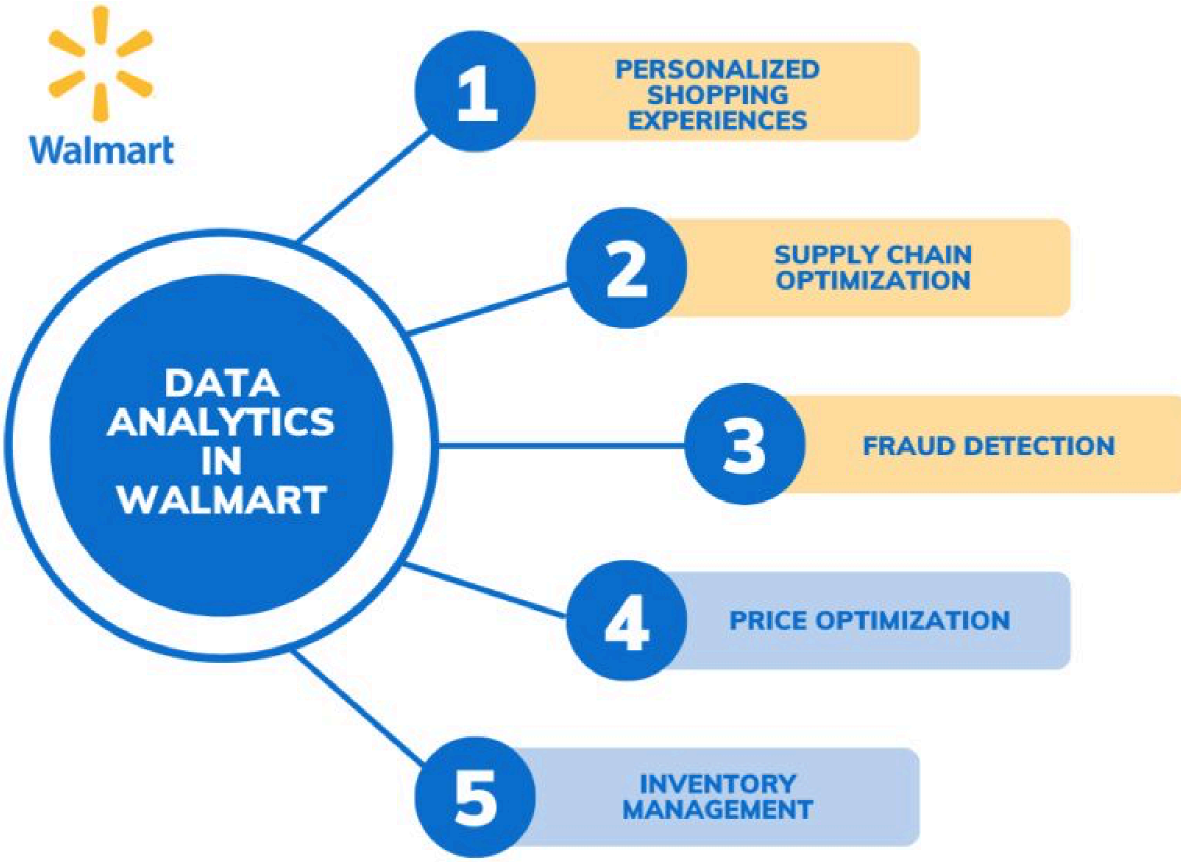
📚 Slide 11: BONUS – Key Data Mining Techniques

Technique	Description	Example
Classification	Categorize into predefined classes	Email Spam Detection
Clustering	Grouping without labels	Customer Segmentation
Association Rules	Relationships among items	Market Basket Analysis
Regression	Predict numeric values	Stock Price Prediction
Anomaly Detection	Outlier detection	Fraud Detection

📊 Slide 12: Real-World Case Study (Walmart)

Problem: Optimize inventory levels
Solution:

- Used **association rule mining** to find product bundles
- Implemented predictive analytics for **seasonal trends**
  Outcome: Reduced stockouts by 23%, improved customer satisfaction



🔚 Slide 13: Summary

- Data Mining uncovers **valuable patterns** from raw data
- Essential in **decision-making** and **predictive modeling**
- Widely used across industries
- Forms the **core intelligence** layer of BI & Analytics systems